

ECE 3340 Numerical Methods

Homework 6: Precision Loss

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Problem 1: Reformulation

known trick for reformulating would be to do $\frac{a^4 + b^4}{a - b}$

Reformulate $f(x) = \underbrace{\sqrt[4]{x+4}}_a - \underbrace{\sqrt[4]{x}}_b$ to mitigate catastrophic cancellation.

① get them out of the root.

$$a^4 - b^4 = (x+4) - x = 4$$

② reformate the eq.

$$a^4 - b^4 = (a-b)(a^3 + a^2b + ab^2 + b^3)$$

$$4 = (a-b)(a^3 + a^2b + ab^2 + b^3) \rightarrow \frac{4}{a^3 + a^2b + ab^2 + b^3} = a - b = f(x) \rightarrow \frac{(x+4) - x}{a^3 + a^2b + ab^2 + b^3} = f(x) \rightarrow f_{new}(x) = \frac{4}{a^3 + a^2b + ab^2 + b^3}$$

$$f_{new} = \frac{4}{a^3 + a^2b + ab^2 + b^3}$$

Before:
 $x=14 \rightarrow (14+4)^{1/4} = 18^{1/4} = 2.04 = A$
 $x^{1/4} \rightarrow 14^{1/4} = 1.93 = B$

$$\frac{(A-B) - 0.125}{0.025} = \frac{0.13 - 0.125}{0.125} = 0.04 \approx 4\%$$

After:

$$\begin{matrix} A^3 = 8.74 \\ A^2B = 8.19 \\ AB^2 = 7.07 \\ B^3 = 7.19 \end{matrix} \rightarrow \frac{4}{31.79} = 0.1258 \approx 0.126$$

$$\downarrow$$

$$\frac{0.126 - 0.125}{0.125} = 0.008 \approx 0.8\%$$

$$f(x) = \frac{4}{(x+4)^{3/4} + (x+4)^{1/4}x^{3/4} + (x+4)^{1/4}x^{1/4} + x^{3/4}}$$

Calculate the relative error before and after using $f(14) = .125$ and 3 digits of precision.

$$E_b = 0.04$$

$$E_a = 0.008$$

Problem 2: Reformulation

Reformulate $f(x) = \frac{1 - \cos x}{2}$ to mitigate catastrophic cancellation.

trig: $1 - \cos x = 2 \sin^2(\frac{x}{2})$

reformate: $f(x) = \frac{2 \sin^2(x/2)}{2} = \sin^2(x/2)$

b) $f(.15) = .561 \times 10^{-2} \approx 0.00561$

Before:

$1 - \cos x$

$$1 - \cos(.15) = 1 - 0.988 = 0.012$$

$$E_b = \frac{0.006 - 0.00561}{0.00561} = .0695$$

After:

$\sin^2(x/2)$

$$\sin^2(\frac{0.15}{2}) \rightarrow \sin(0.075)^2 = 0.000...17$$

$$\left| \frac{0.000...17 - 0.00561}{0.00561} \right| \rightarrow 0.999 \approx 1$$

← technically this value is a negative

$$f(x) = \sin^2(x/2)$$

Calculate the relative error before and after using $f(.15) = .561 \times 10^{-2}$ and 3 digits of precision.

$$E_b = 0.0695$$

$$E_a = 1.00$$

Problem 3: Maclaurin Series

Use a 2nd-order Maclaurin series to reformulate $f(x) = e^x - \cos(x)$ and mitigate catastrophic cancellation.

Note: we stop at $\frac{x^2}{2}$ cause we only need the 2nd order

$f(x) = e^x - \cos x \rightarrow$ 2nd MCS $e^x \approx 1 + x + \frac{x^2}{2}$
 $\rightarrow \cos x \approx 1 - \frac{x^2}{2}$
 $e^x - \cos x = (1 + x + \frac{x^2}{2}) - (1 - \frac{x^2}{2}) = x + x^2 = f_n$
 $\frac{x^2}{2} + \frac{x^2}{2} = x^2$

$$f(x) = x + x^2$$

Before: $e^{0.123} \approx 1.131237 \rightarrow 1 + 0.1237 + \frac{(0.1237)^2}{2} = 1.01238 \approx 1.01$
 $\cos(0.123) \approx 0.999924 \rightarrow 1 - \frac{(0.099924)^2}{2} = 0.999924 \approx 1.00$
 $E_n = 1.01 - 1.00 = 0.01 \rightarrow \frac{0.01 - 0.0125}{0.0125} = 0.20 \times 100 = 20\%$

Calculate the relative error before and after using $f(0.0123) = 0.0125$ and 3 digits of precision.

After: $x + x^2 = 0.0123 + 0.00015129 \approx 0.01245129 \approx 0.0125$
 $\frac{0.0125 - 0.0125}{0.0125} \times 100 = 0\%$

$E_b = 0.20$	$E_a = 0.01$
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Problem 4: Mean Value Theorem

Use the Mean Value Theorem to reformulate $f(x) = \sqrt{100} - \sqrt{101}$ and mitigate catastrophic cancellation.

$\sqrt{100} = 10$
 $\sqrt{101} = 10.04987$
 $f_{true} = 10 - 10.04987 = -0.04987 \approx 0.04987$
 Before:
 $\sqrt{100} = 10$
 $\sqrt{101} = 10.04987 \approx 10.0$
 $\frac{f_n - f_{true}}{f_{true}} \rightarrow \frac{0.0 - 0.04987}{0.04987} = 1.0 \times 100 = 100\%$

After: $\frac{-1}{10 + \sqrt{101}} \rightarrow \frac{-1}{20} = -0.05$
 $\frac{f_n - f_{true}}{f_{true}} \rightarrow \frac{-0.05 - (-0.04987)}{0.04987} = \frac{0.0001294}{0.04987} = 0.002595 \times 100 \approx 0.26\%$

$$f(x) = \frac{-1}{10 + \sqrt{101}}$$

Calculate the relative error before and after using 3 digits of precision.

$E_b = 1.00$	$E_a = 0.002$
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